

5.1 Dataset Used

The dataset used in this project is a specialized collection of images of banana plant leaves, designed to detect various diseases that affect banana crops. The images were self-collected, primarily from local farms and agricultural sources, ensuring that they accurately represent real-world conditions. This dataset is hosted in Google Drive and is organized into three main subdirectories to facilitate training, validation, and testing of machine learning models. The distribution of data is as follows:

- **Training Set (70%):** This portion of the dataset is used to train the models. It contains a diverse set of images that cover all the classes. The model learns to identify patterns and features within the images, such as leaf texture, color, and shape, which are associated with the different diseases or healthy conditions.
- **Validation Set (15%):** This set is used to tune the model and adjust hyperparameters. It helps in evaluating the model's performance during the training process and ensures that it doesn't overfit to the training data. This set is used to periodically assess the model's accuracy and generalization ability.
- **Test Set (15%):** The test set is used for final evaluation of the model. It provides an unbiased performance assessment after the model has been trained and tuned. The results on the test set give an estimate of how the model would perform in real-world applications.

The dataset contains five primary classes, each representing a different condition that the banana plant can experience:

1. **Black_sigatoka:** This class contains images of banana leaves infected with the black Sigatoka fungus, a common disease that causes dark streaks and spots on the leaves, eventually affecting the plant's growth.
2. **Yellow_sigatoka:** Similar to black Sigatoka, this class represents banana leaves infected with yellow Sigatoka, which causes yellowing of the leaves and can lead to premature leaf death, reducing the plant's productivity.
3. **Healthy:** This class contains images of healthy banana leaves, with no visible signs of disease. These images are important for helping the model differentiate between diseased and healthy plants.
4. **Potassium_deficiency:** This class consists of images of banana leaves showing signs of potassium deficiency, which may appear as yellowing at the edges of the leaves and can severely affect plant growth and fruit production.
5. **Panama_disease:** This class represents images of banana plants infected with Panama disease, a soil-borne fungal infection that causes wilting and yellowing of the leaves. It is a major threat to banana plantations worldwide.

The dataset is structured in such a way that each of the classes contains a varying number of images. The images are captured under different lighting conditions, from various angles, and may include variations in leaf damage. This diversity in the dataset allows the model to generalize well to real-world conditions, improving its robustness and accuracy.

The dataset provides a balanced representation of these five conditions, which is crucial for training a multi-class classification model. A balanced dataset ensures that the model is not biased toward any particular class and that it has enough data to learn from each category equally. This is essential for achieving high performance across all classes, especially when dealing with diseases that may not have as many visual features or may be subtler than others.

By using this structured and diverse dataset, the model is better equipped to classify banana leaves into the correct disease or healthy category, making it a valuable tool for automated disease detection in banana plantations.

5.2 Performance Metrics

The following metrics were used to evaluate the performance of the models:

- **Accuracy:** The percentage of correct predictions made by the model out of all predictions.
- **Precision:** The proportion of true positive predictions in relation to the total predicted positives (how many predicted as a particular class are actually that class).
- **Recall:** The proportion of true positives in relation to the actual positives (how many of the actual positives were predicted correctly).
- **F1-score:** The harmonic mean of precision and recall, providing a balance between the two, especially useful when the class distribution is imbalanced.

- **Confusion Matrix:** A table that summarizes the performance of the model by showing the actual versus predicted class labels.

These metrics are suitable for multi-class classification because they allow for detailed insight into how well each class is predicted and where the model might be making errors. Using accuracy alone may not be enough in multi-class scenarios, which is why precision, recall, and F1-score are included to provide a more comprehensive view of model performance.

5.3 Model Evaluation

For each model, the following performance details will be provided:

- **Model 1: CNN**

Fig.5.3.1 Confusion Matrix(CNN)

Fig.5.3.2 Classification Report(CNN)

Fig.5.3.3 Test Accuracy(CNN)

- **Model 2: EfficientNet**

Fig.5.3.1 Confusion Matrix(EfficientNet)

Fig.5.3.2 Classification Report(EfficientNet)

Fig.5.3.3 Test Accuracy(EfficientNet)

- **Model 3: MobileNetV2**

Fig.5.3.1 Confusion Matrix(MobileNetV2)

Fig.5.3.2 Classification Report(MobileNetV2)

Fig.5.3.3 Test Accuracy(MobileNetV2)

- **Model 4: DenseNet**

Fig.5.3.1 Confusion Matrix(DenseNet)

Fig.5.3.2 Classification Report(DenseNet)

Fig.5.3.3 Test Accuracy(DenseNet)

- **Model 5: InceptionV3**

Fig.5.3.1 Confusion Matrix(InceptionV3)

Fig.5.3.2 Classification Report(InceptionV3)

Fig.5.3.3 Test Accuracy(InceptionV3)

- **Model 6: ResNeXt**

Fig.5.3.1 Confusion Matrix(ResNeXt)

Fig.5.3.2 Classification Report(ResNeXt)

Fig.5.3.3 Test Accuracy(ResNeXt)

The evaluation results presented here are based on Semester VI work. Further improvements, deployment strategies, and extended analysis will be undertaken in Semester VII.